

Modelling Particle Production in Analog (3+1) Inflationary Cosmologies Using High-Performance Computing

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Introduction - Analog Universes

1 Inflation created particles

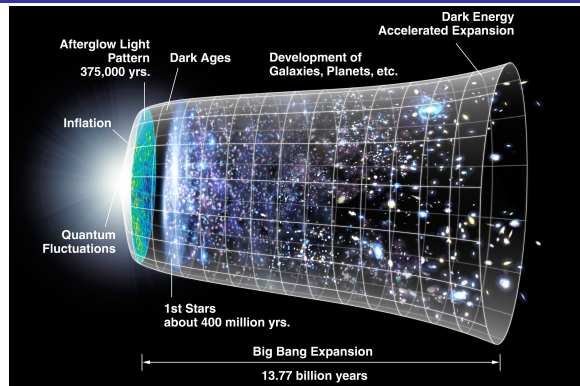


Figure: History of the universe highlighting period of rapid expansion

Source: https://upload.wikimedia.org/wikipedia/commons/6/6f/CMB_Timeline300_no_WMAP.jpg

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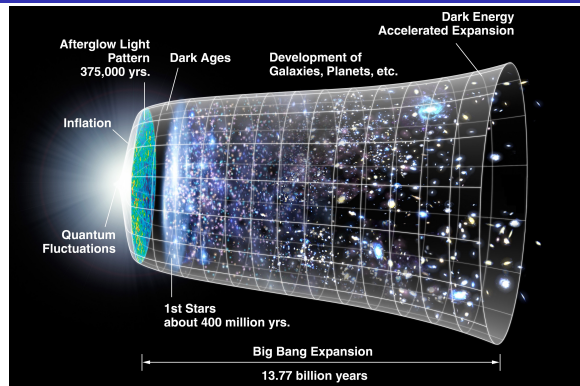


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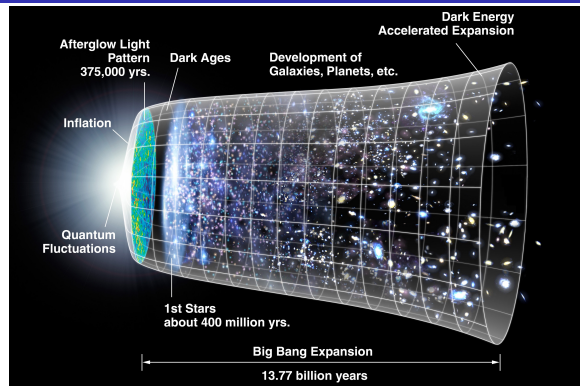


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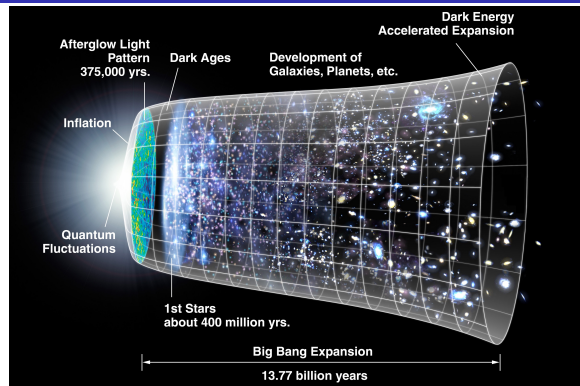


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- 5 Lab based system = “analog universe”

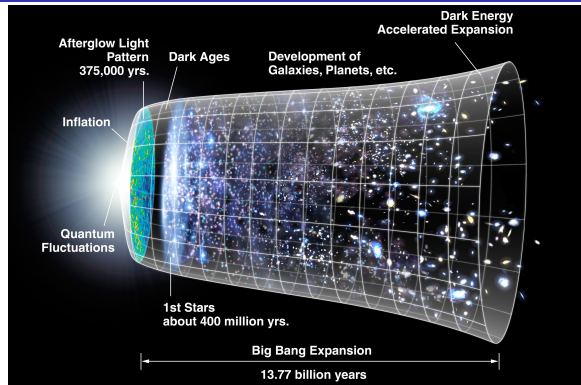


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Introduction - The Story So Far

- Many investigations into analog universes
e.g. Visser and Wienfurtner



Figure: Subsets of increasingly specific literature

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- Few investigations into exact analogs



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- Few investigations into exact analogs
- Fewer into computational simulation
- Even fewer using high-performance computing



Figure: Subsets of increasingly specific literature

Methods - The Field Equation

- One equation describes both the phase of ultra-cold quantum gases and inflation (a) particles

The Field Equation

$$\partial_t^2 \tilde{\theta}_{\mathbf{k}} - \frac{\dot{a}}{a} \partial_t \tilde{\theta}_{\mathbf{k}} - ac_0^2 k^2 \tilde{\theta}_{\mathbf{k}} = 0$$

$$\tilde{\theta}_{\mathbf{k}}(0) = \tilde{\theta}_{\mathbf{k}0} \quad \partial_t \tilde{\theta}_{\mathbf{k}}(0) = -U_0 n_{\mathbf{k}0} / \hbar$$

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- One equation for each $k < k_c(t)$ the trans-phononic cutoff

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Methods - Symplectic Integrators

- Minimize energy error

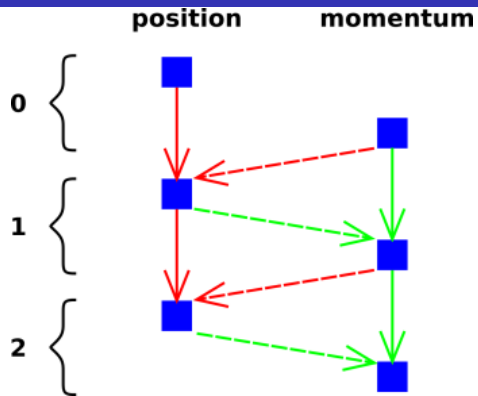


Figure: Symplectic integrator example: Velocity Verlet

Source: <https://www.av8n.com/physics/symplectic-integrator.htm>

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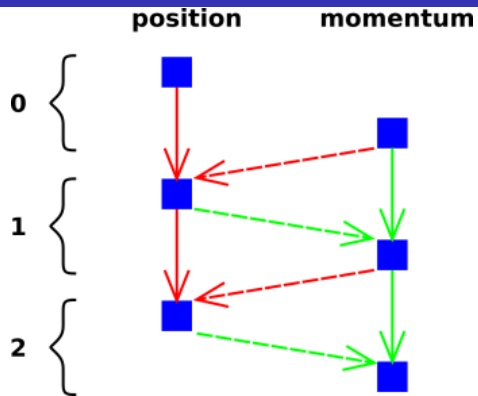


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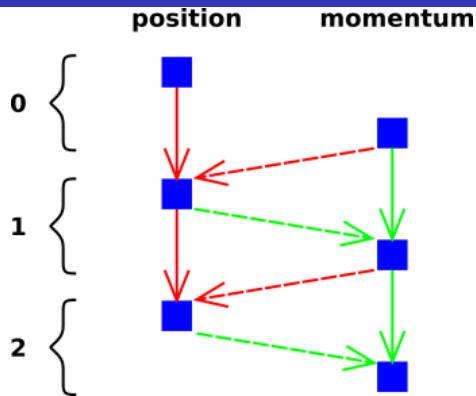


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- Minimize energy error
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- Use new momentum to calculate position
- More complex but more stable

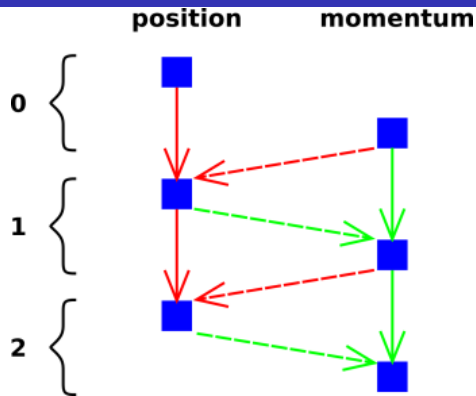


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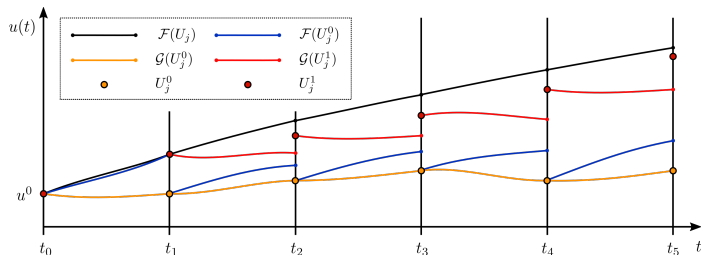


Figure: Illustration of the first iteration in Parareal

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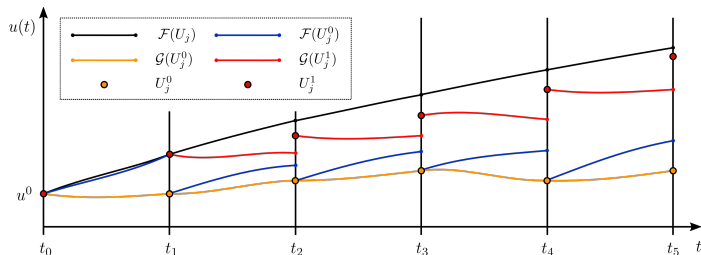


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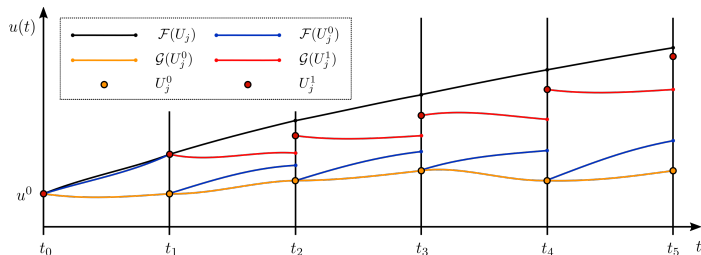


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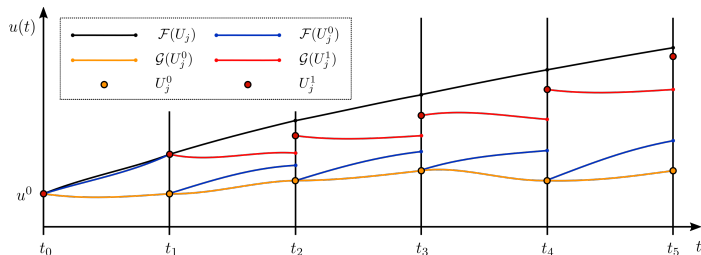


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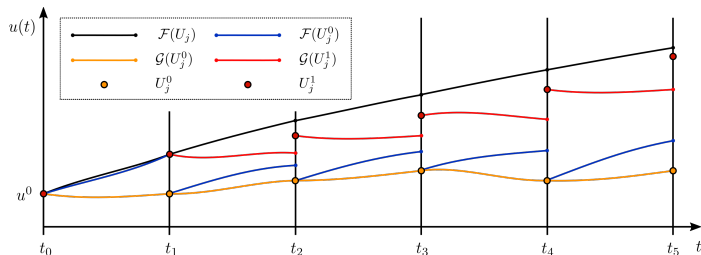


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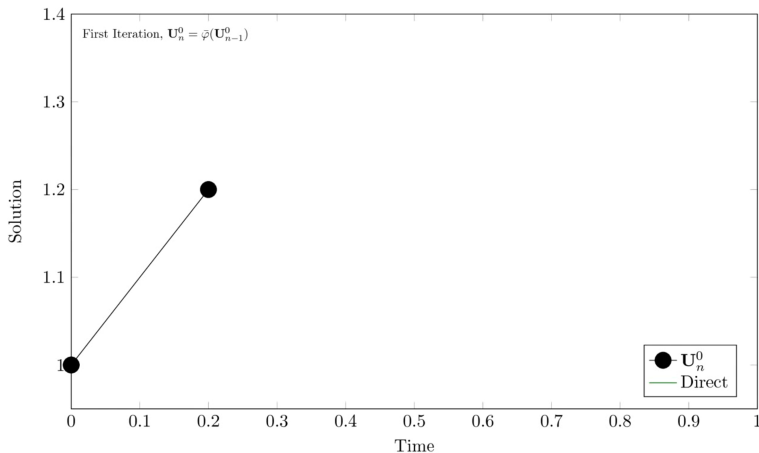


Figure: The *Parareal* approach to solving initial value problems in parallel.
Coarse: $\bar{\varphi}$, Fine: φ

Methods - High-Performance Computing

- Equation for each $k < k_c$

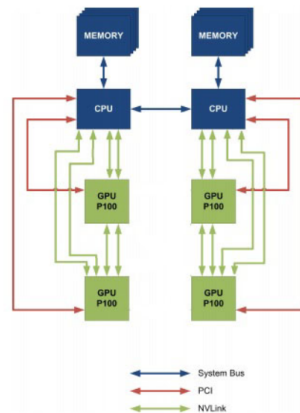


Figure: CPU to GPU and GPU to GPU interconnect using NVLink

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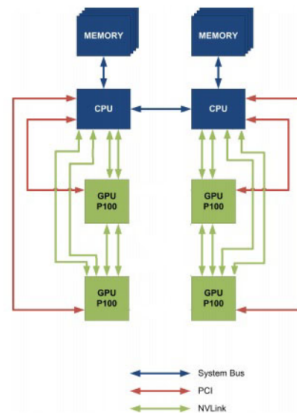


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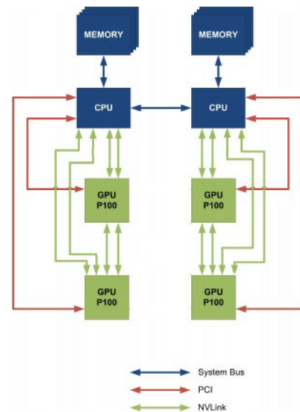


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Methods - High-Performance Computing

- Equation for each $k < k_c$
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- Use CWU's "Turing" supercomputer
- 12 NVIDIA P100 GPUs
 - Each 83% performance of modern RTX 3060

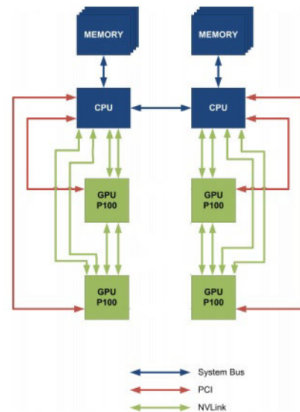


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Analytic Results - Simple Expansion

- Simple expansion: $a(t) = e^{-t/t_s}$

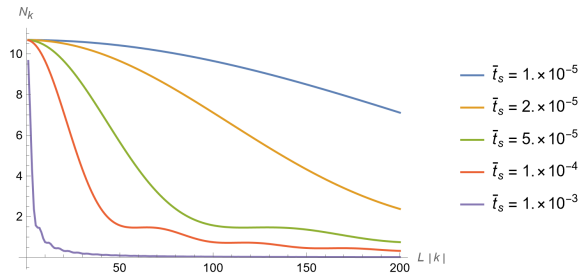


Figure: The momentum distribution when the universe stops expanding for multiple effective rates of expansion

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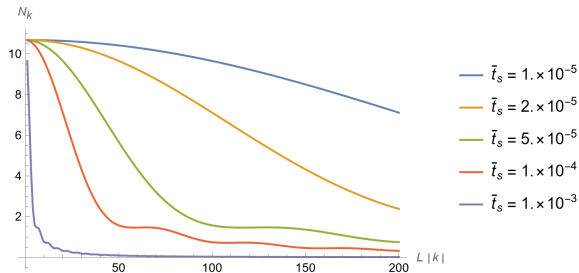


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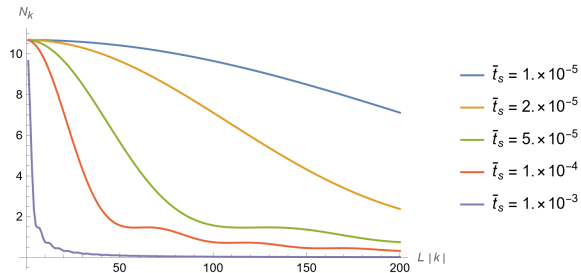


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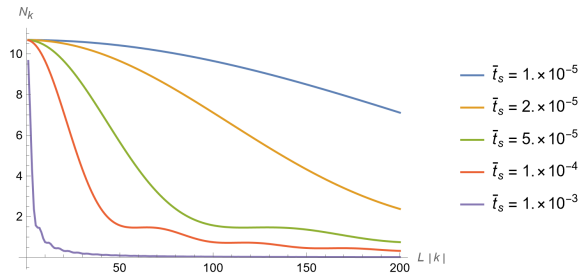


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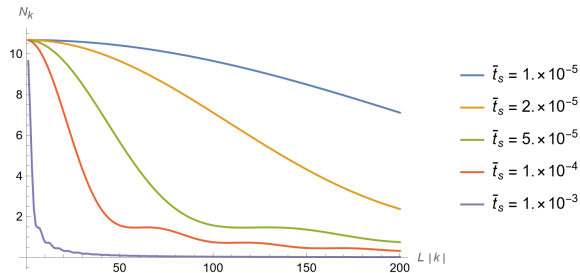


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Acknowledgements

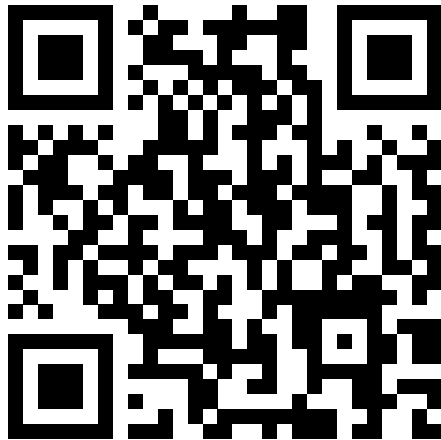
- Dr. Andy Piacsek (CWU, Physics)
- Dr. Szilard VAJDA (CWU, Computer Science)
- Dr. Micheal Braunstein (CWU, Physics)
- Dr. Brandon Peden (WWU, Physics & Astronomy)

Thank you & Code source

Thank you for your attention.

The code for this project is available at:

- GitHub (QR code): github.com/nondairyneutrino/thesis
- My website: nathanwchapman.com



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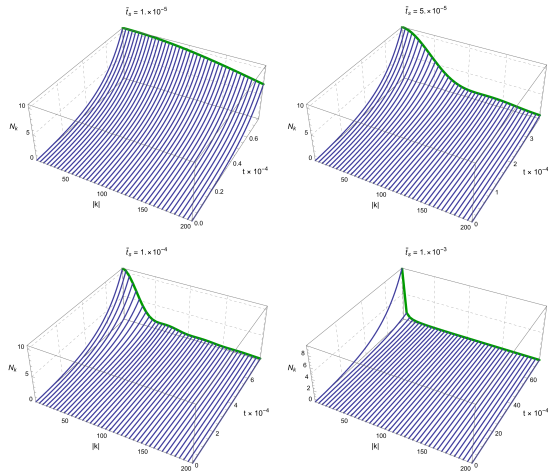


Figure: Momentum distribution throughout the expansion